Problem 4

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Problem 1 *Two polynomials, h and g, are given*

$$h(x) = \frac{x^2 - 4}{4}$$

$$g(x) = x - 2$$

State the form of the limit, evaluate the limit, or state that it does not exist. Justify your answer.

(a) $\lim x \to 2 \frac{h(x)}{g(x)}$

Explanation. $\lim x \to 2 \frac{h(x)}{g(x)} = \lim x \to 2 \frac{\frac{x^2-4}{4}}{x-2}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim x \to 2 \frac{\frac{x^2-4}{4}}{x-2} &= \lim x \to 2 \frac{(x-2)(x+2)}{4(x-2)} \\ &= \lim x \to 2 \frac{x+2}{4} \\ &= \frac{4}{4} \\ &= 1 \end{aligned}$$

(b) $\lim x \to 3 \frac{g(x) - g(3)}{x-3}$

Explanation. $\lim x \to 3 \frac{g(x) - g(3)}{x-3} = \lim x \to 3 \frac{x-2-1}{x-3}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim x \to 3 \frac{g(x) - g(3)}{x-3} &= \lim x \to 3 \frac{x-2-1}{x-3} \\ &= \lim x \to 31 \\ &= 1 \end{aligned}$$

(c) $\lim x \to 4 \frac{h(x) - h(4)}{x-4}$

Explanation. $\lim x \to 4 \dfrac{h(x) - h(4)}{x - 4} = \lim x \to 4 \dfrac{\frac{x^2-4}{4} - 3}{x - 4}$
Form: $\dfrac{0}{0}$

$$\begin{aligned}
\lim x \to 4 \frac{h(x) - h(4)}{x - 4} &= \lim x \to 4 \frac{\frac{x^2-4}{4} - 3}{x - 4} \\
&= \lim x \to 4 \frac{x^2 - 4 - 12}{4(x-4)} \\
&= \lim x \to 4 \frac{x^2 - 16}{4(x-4)} \\
&= \lim x \to 4 \frac{(x-4)(x+4)}{4(x-4)} \\
&= \lim x \to 4 \frac{x+4}{4} \\
&= 2
\end{aligned}$$